CPEN 416

Lecture 10: Quantum compilation I

Thursday 15 October 2026

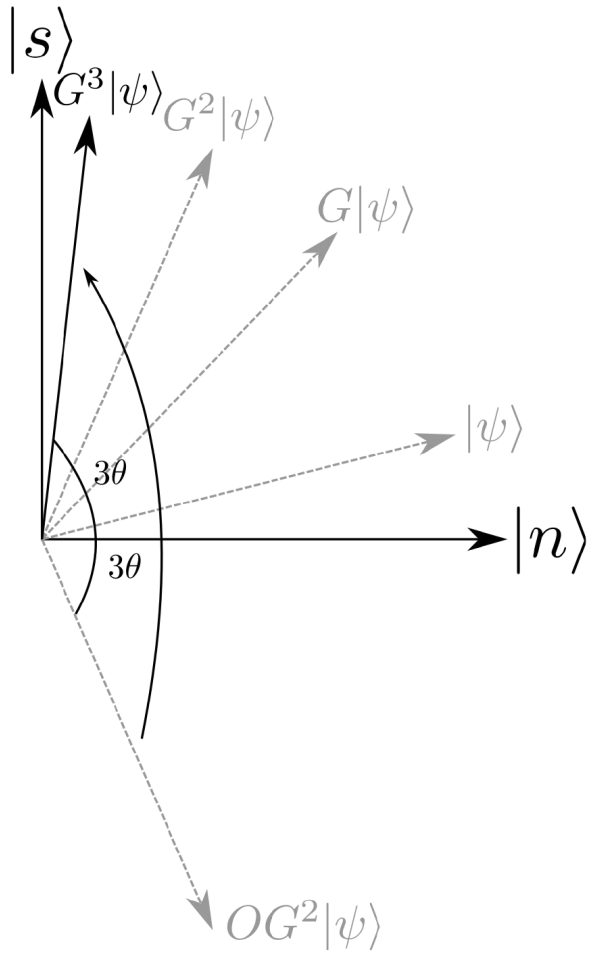
Announcements

- Project details available by tomorrow
- *No formal tutorial Monday*, but Parvin will hold office hours in usual room
- Quiz 5 on Tuesday

Last time

We developed a geometric formulation for Grover's algorithm as a series of reflections implemented by the oracle and diffusion operators.

Grover's algorithm yields a square-root speedup in **query complexity** over classical methods for unstructured search.

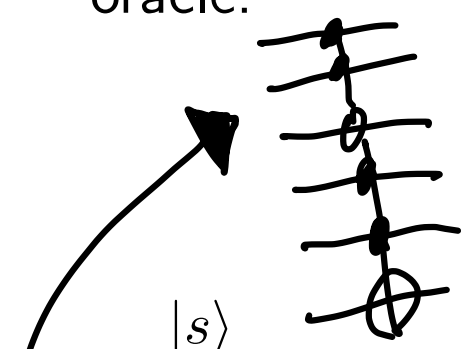


Learning outcomes

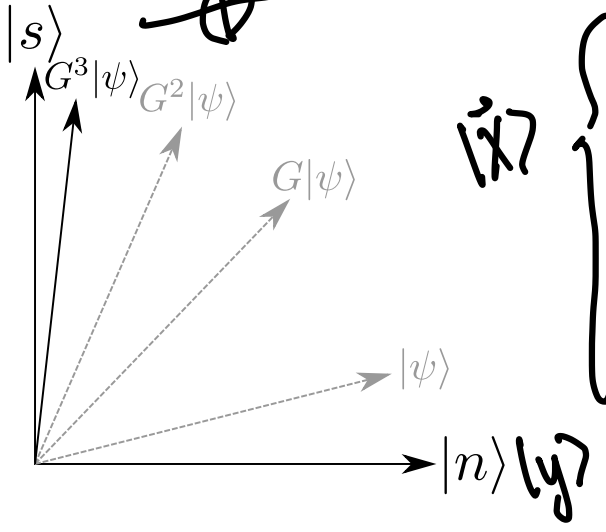
- implement Grover's algorithm in PennyLane
- define and calculate cost metrics of quantum circuits
- define and list key universal gate sets
- decompose selected single- and multi-qubit operations using one- and two-qubit universal gate sets

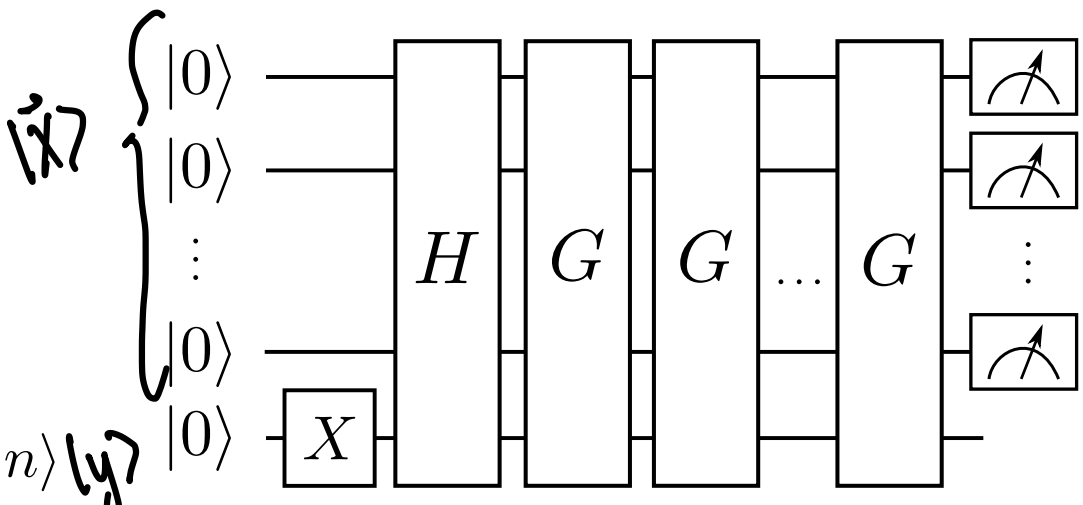
Implementing Grover search

We need to design oracle and diffusion circuits. Use the "type 1" oracle:



$O|\vec{x}\rangle|y\rangle = |\vec{x}\rangle|y \oplus f(\vec{x})\rangle$



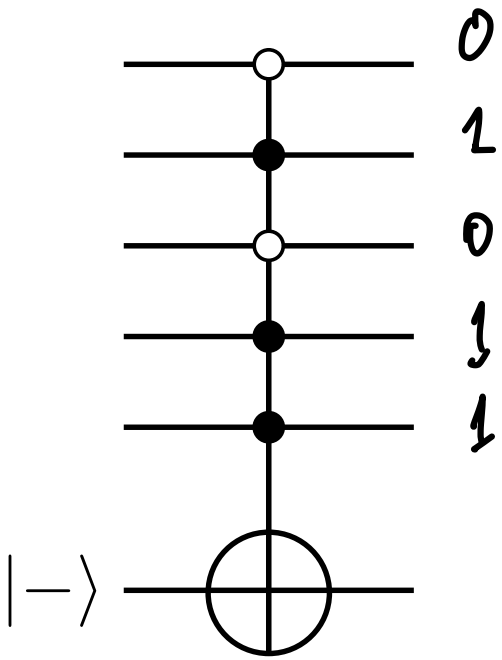


Solution $|\vec{s}\rangle \longrightarrow O|\vec{s}\rangle|y\rangle = |\vec{s}\rangle|y \oplus 1\rangle$

$|\vec{s}\rangle = |1\,1\,0\,1\,1\rangle$

The oracle circuit

Exercise: show that a multicontrolled X gate, controlled on **s**, can be used as an oracle:

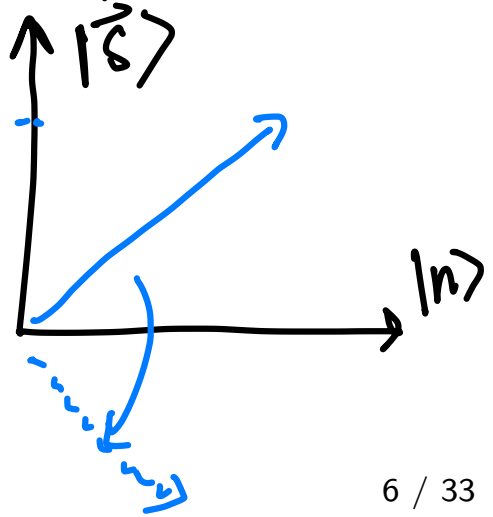


$$O\left[\,|10110\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\,\right]$$

$$= |10110\rangle \otimes \frac{1}{\sqrt{2}}(|1\rangle - |0\rangle)$$

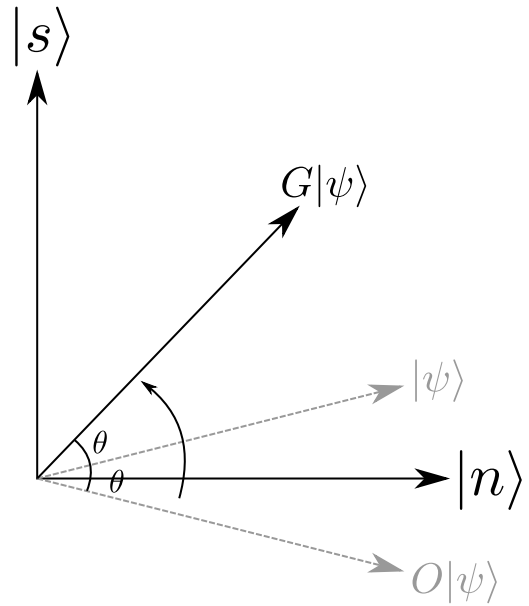
$$= -|10110\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$= (-1)\,|\vec{s}\rangle|-\rangle$$



The diffusion circuit

The diffusion operator performs a reflection about the uniform superposition, $|\psi\rangle$.



The diffusion circuit

The unitary matrix given by

$$D = 2|\psi\rangle\langle\psi| - I$$

is equivalent to the diffusion operator.

$|\psi\rangle = H^{\otimes n}|0\rangle^{\otimes n}$

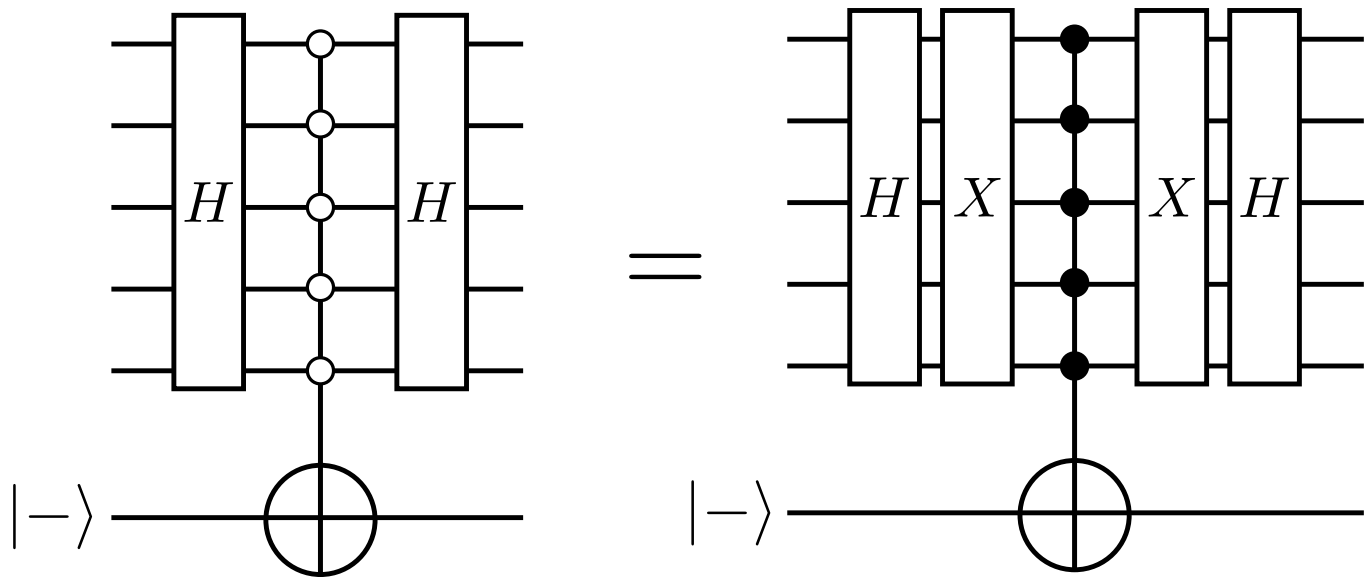
$D = 2H^{\otimes n}|0\rangle^{\otimes n}\langle 0|^{\otimes n}H^{\otimes n} - I$

$= 2H^{\otimes n}|0\rangle^{\otimes n}\langle 0|^{\otimes n}H^{\otimes n} - H^{\otimes n}\cdot H^{\otimes n}$

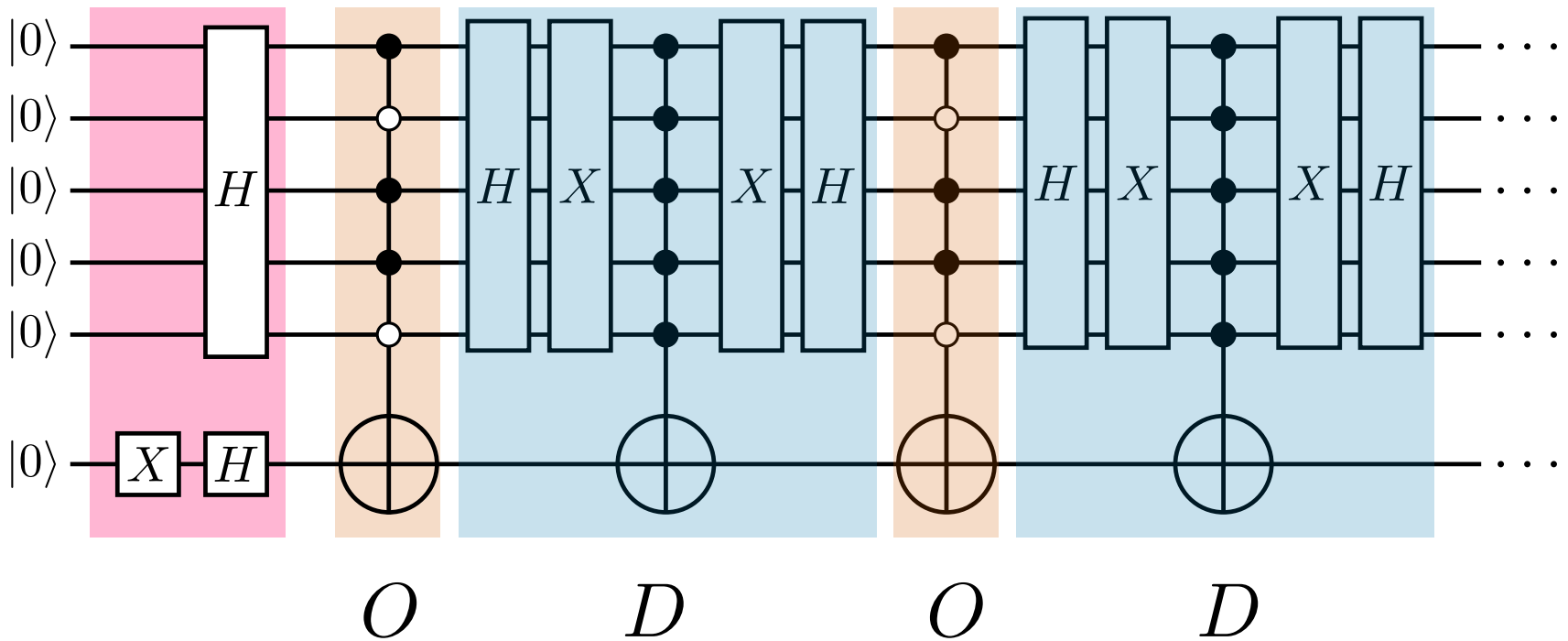
$= H^{\otimes n}\left[2|0\rangle^{\otimes n}\langle 0|^{\otimes n} - I\right]H^{\otimes n}$

$2|00\rangle\langle 00| - I = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = -\begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$

The diffusion circuit

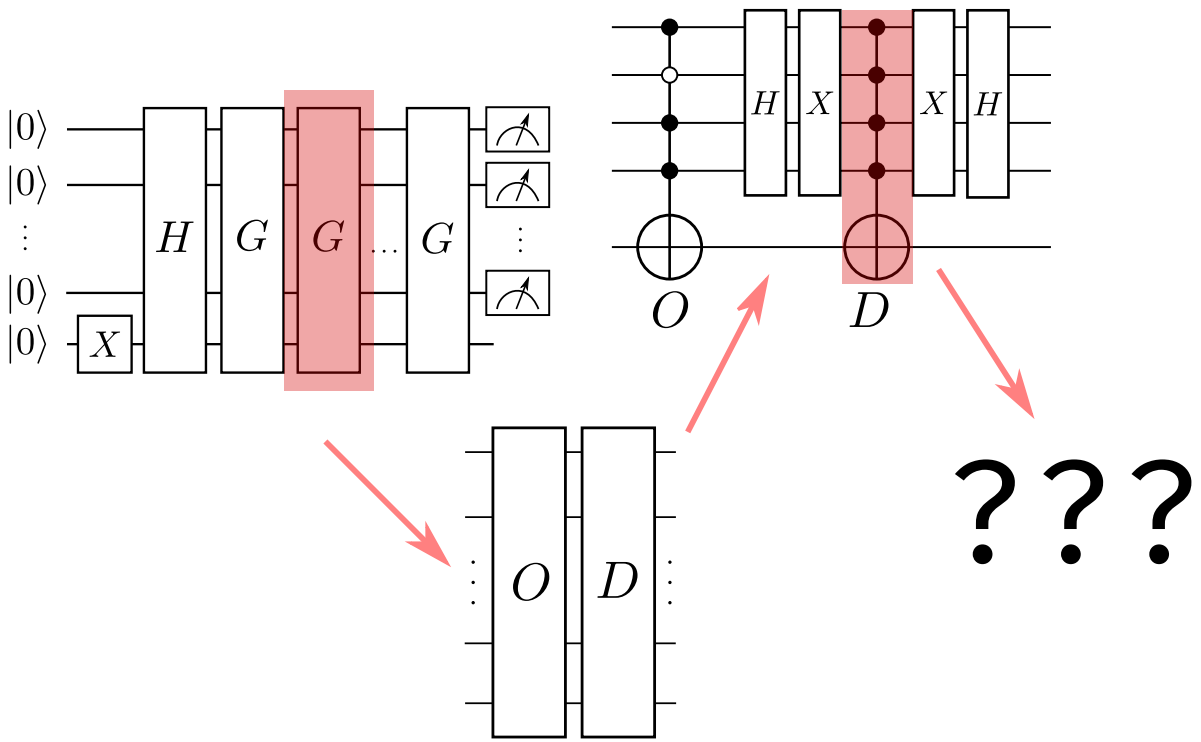


The full Grover circuit



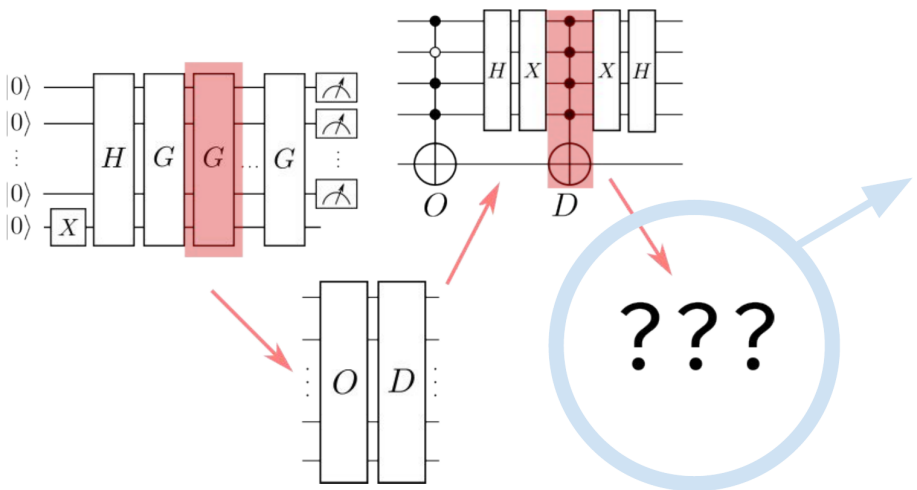
The full Grover circuit

Clearly, each of the $O(\sqrt{2^n})$ queries requires some number of gates... how much does Grover *really* cost?



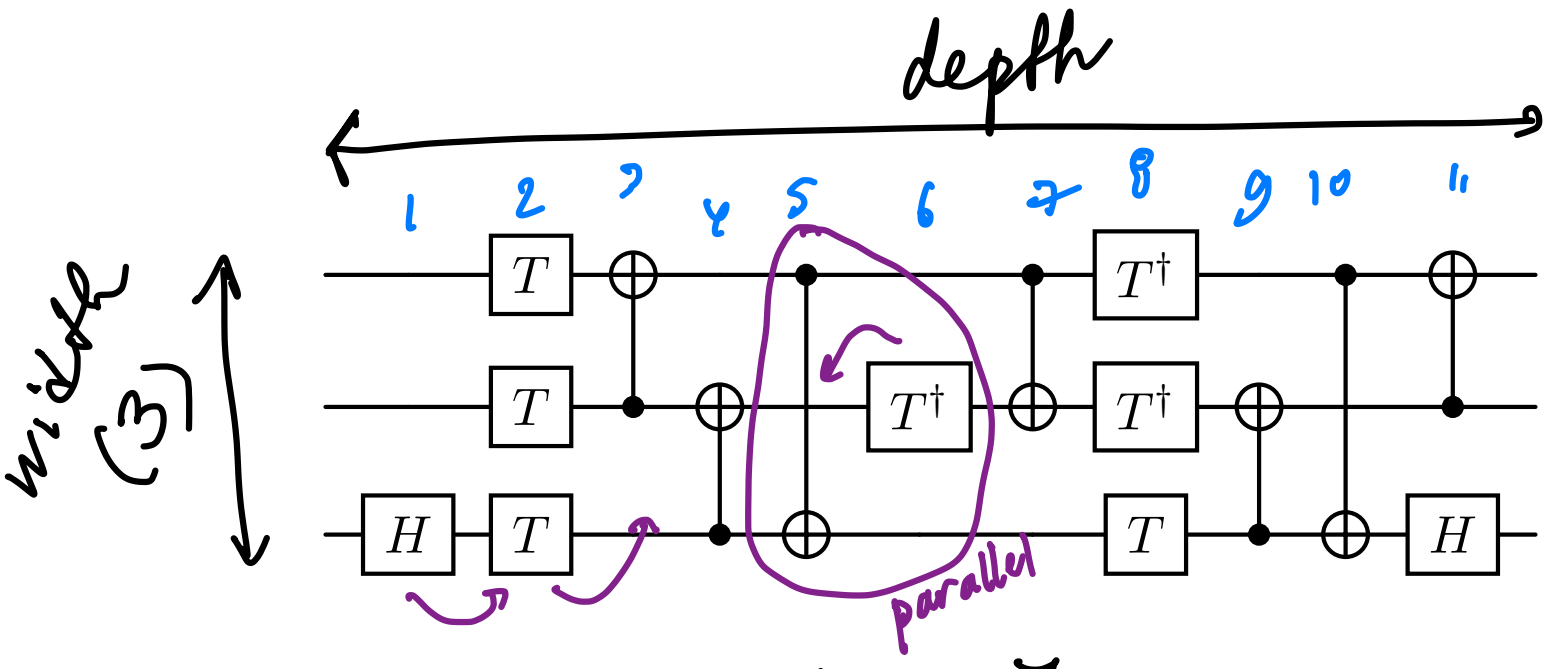
Let's open the black box!

Quantum circuit synthesis and optimization



- How do we decide which gates to use?
- What criteria determine whether our choice is "good"?
- How do we improve on our choices?

Width, depth, gate count



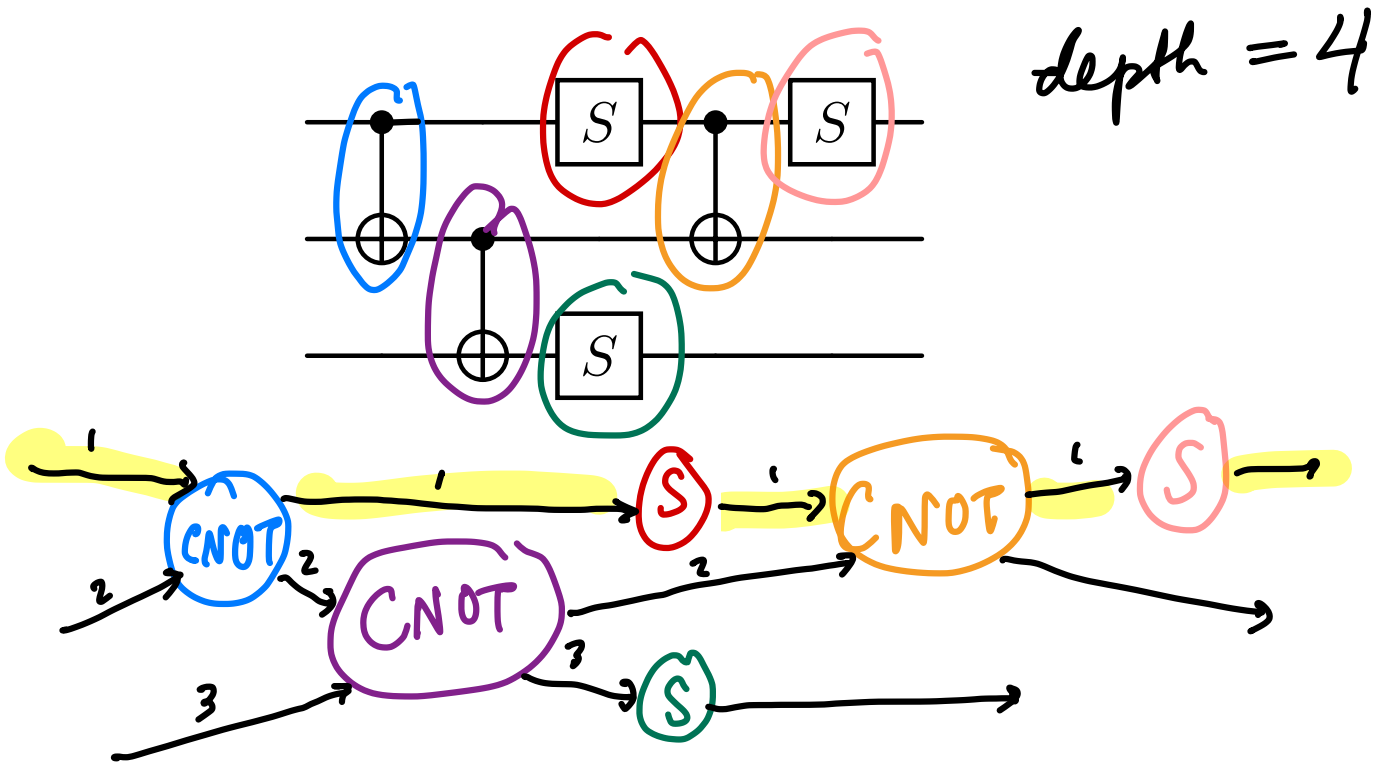
→ depth 9 w/ parallelization

CNOT gates: 7 (2 qubit)

One qubit gates: 9
- T count 7
- H count 2

Circuit depth

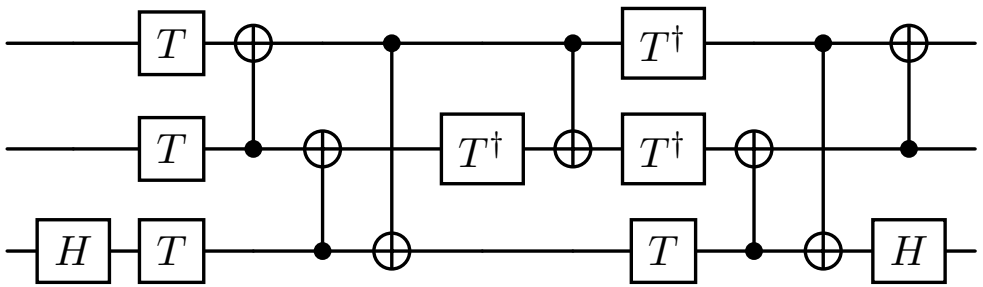
The *circuit depth* is the length of the longest path in its **directed acyclic graph (DAG)** representation.

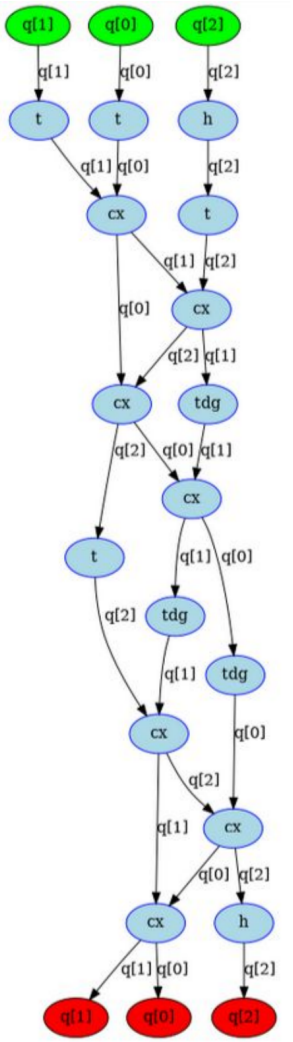


Circuit depth

Generally don't do by hand; NetworkX is your best friend.

`https://networkx.org/`

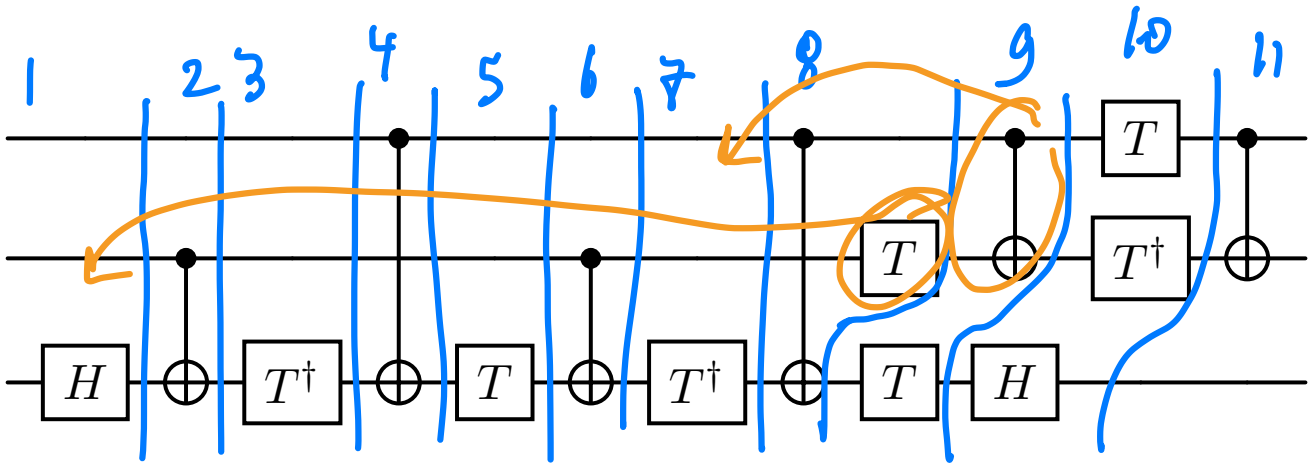




(DAG generated in Qiskit)

Exercise

Determine resource counts of the following circuit.



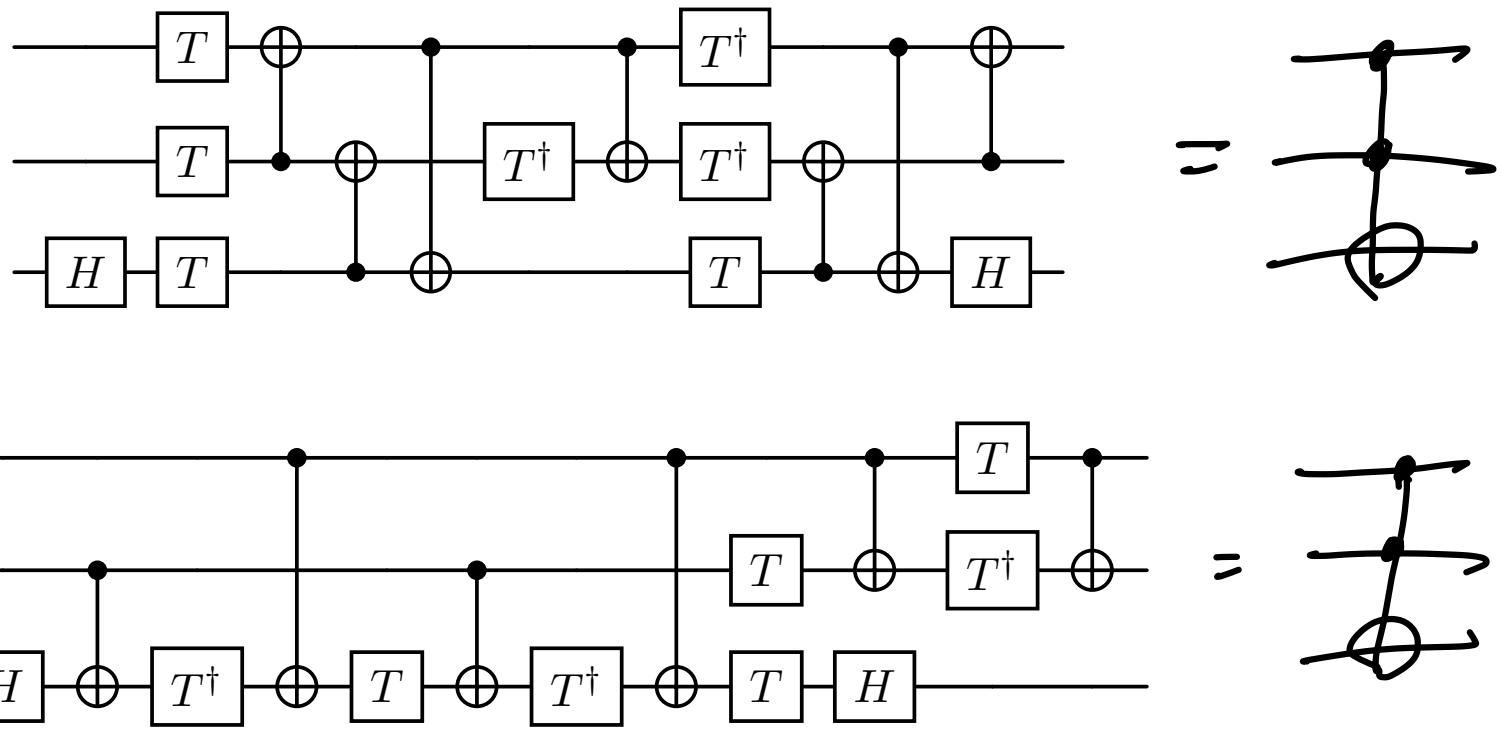
width = 3

depth 12, 10, 11 ??

CNOT: 6 T: 7 H: 2

Circuit synthesis and decomposition

What do these circuits do?



Circuit synthesis

Given:

- an arbitrary unitary matrix, U
- a finite set of gates, $\{G_k\}$
- a target precision, ϵ

Find

$$U' = C_1 C_2 \cdots C_m$$

such that

$$\|U - U'\| < \epsilon, \quad C_i \in \{G_k\}$$

Universal gate sets

A $\{G_k\}$ for which this can be done is called a **universal gate set**.

Examples for a single qubit:

- Any two Pauli rotations (exact)
- Hadamard and T (approximate)

HTTTHTTHT···

Theoretical bound: $O(\log^c(1/\epsilon))$ gates, c varies from 1-4 depending on assumptions.

Universal gate sets

Using a different gate set is like *speaking a different language*.



I would like to use a real quantum computer.

I would like to use a real quantum computer.



J'aimerais utiliser un vrai ordinateur quantique.

I would like to use a real computer that is quantum.



本当の量子コンピュータを使いたい

A real quantum computer, I would like to use.

Universal gate sets

Any single-qubit unitary can be expressed using a sequence of 3 Pauli rotations around a minimum of 2 unique axes

- *ZYZ*
- *XYX*
- *XYZ*
- 9 other combinations (*Euler angles*)

qml.Rot

class Rot(*phi, theta, omega, wires, id=None*)

Bases: `pennylane.operation.Operation`

Arbitrary single qubit rotation

$$R(\phi,\theta,\omega) = RZ(\omega)RY(\theta)RZ(\phi) = \begin{bmatrix} e^{-i(\phi+\omega)/2}\cos(\theta/2) & -e^{i(\phi-\omega)/2}\sin(\theta/2) \\ e^{-i(\phi-\omega)/2}\sin(\theta/2) & e^{i(\phi+\omega)/2}\cos(\theta/2) \end{bmatrix}.$$

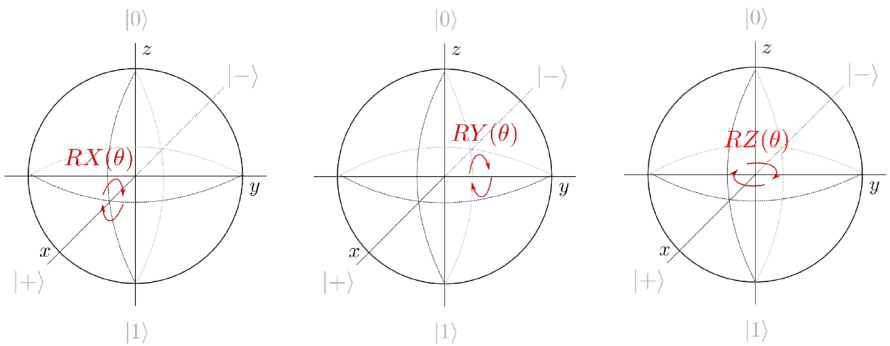
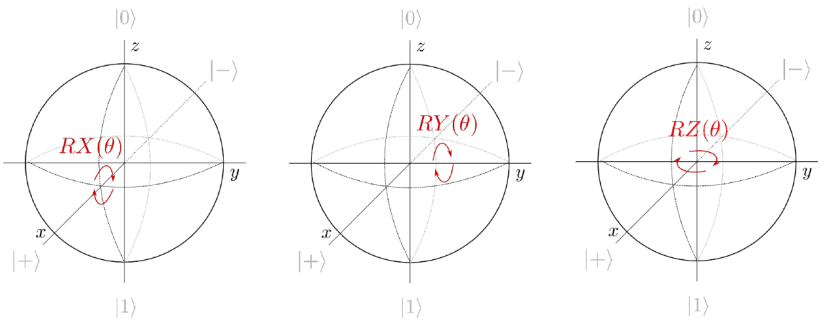


Image credit: Codebook node I.6

Hardware-specific example

RX/*RY*/*RZ* are **arbitrary-angle** rotations about **fixed axes**



IonQ's trapped ion hardware native gates: *GPI*/*GPI*2, **fixed-angle** rotations about **arbitrary axes** in *xy*-plane

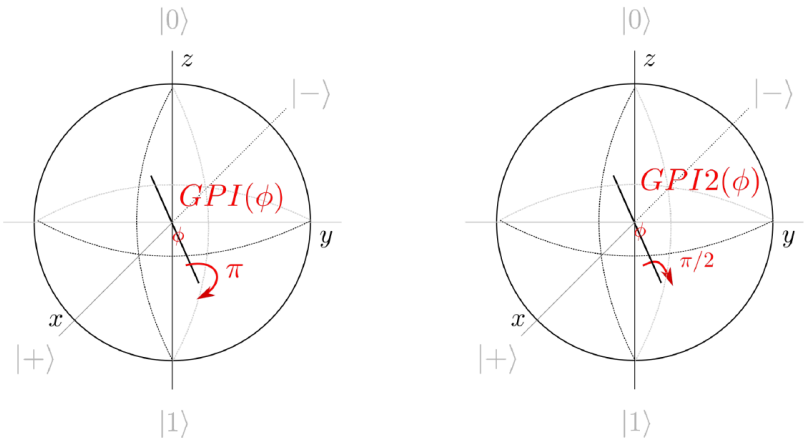


Image credit: adapted from Codebook node I.6.

See blog post: `https://pennylane.ai/blog/2023/06/the-ionizer-building-a-hardware-specific-transpiler-using-pennylane`

Next time

Content:

- The quantum compilation stack (from top to bottom)
- Quantum circuit optimization

Action items:

1. Prepare for Quiz 5
2. Watch PrairieLearn for project information

Recommended reading:

- Nielsen & Chuang 4.5
- Compiling quantum circuits (PennyLane documentation)
- Explore the `qp.transforms` module